

Arslan Khalid

Python Developer

iarslankhalid@yahoo.com

[LinkedIn](#)

[GitHub](#)

[WhatsApp](#)

[\(+92\) 306-6200905](#)

EDUCATION

B.Sc. Electrical Engineering

University of Gujrat

CGPA: 3.12

Nov 2020 – Jul 2024

Relevant Courses:

- [Advanced ML Algorithms by Stanford University](#)
- [Supervised Machine Learning by Stanford University](#)
- [Django Web Framework by Meta](#)

SKILLS

Languages:	Python · C++
Libraries/Frameworks:	BeautifulSoup · FastAPI · Django · TensorFlow · Hugging Face
Machine Learning:	NLP · Computer Vision · Reinforcement Learning (RL) · RAG Models
Version Control:	Git · GitHub · GitLab
Development:	Clean Code · Deploying ML Models · Python for Automation
Soft Skills:	Team Player · Problem Solving · Critical Thinking · Communication Skills

EXPERIENCE

- GulzarSoft (On-site)** Gujrat, Pakistan
Python Developer Oct 2024 – Present
 - Car Recommendation System:** Designed and implemented a collaborative filtering model for a car recommendation system, enhancing recommendation accuracy by 30%.
- Coders Launchpad (Remote)** Dubai, UAE
Machine Learning Intern Nov 2023 – Feb 2024
 - Voice-to-Voice Conversion Model:** Assisted in training a voice-to-voice conversion model for the Quran Project using RVC, contributing to high-quality audio synthesis.
 - Audio Synthesis:** Enhanced audio quality for Quran recitations, improving accuracy and naturalness in voice conversion.
- Intelligent Systems Lab (On-site)** Gujrat, Pakistan
Machine Learning Intern Sep 2023 – Oct 2023
 - Facial Recognition Model:** Developed a facial recognition model using TensorFlow to match ID card images with live photos, improving identification accuracy.
 - Image Matching:** Leveraged machine learning techniques to accurately match facial features between static and live images, streamlining identification.

PROJECTS

- ZARM — Backend & AI Integration System**
FastAPI, PostgreSQL, Redis, OpenAI
 - Engineered a high-performance, asynchronous backend architecture utilizing FastAPI, PostgreSQL, and Redis to drive scalable consulting workflows.
 - Architected secure, multi-tenant database models to ensure strict data isolation and efficient querying across multiple client environments.
 - Integrated advanced AI agent layers into the backend infrastructure to automate data synthesis and client reporting, reducing manual processing time by 85%.
- Jarvis — Real-Time AI Voice Assistant**
LiveKit, Python, WebSockets, Wake-Word Detection
 - Architected a persistent, low-latency AI voice assistant leveraging LiveKit real-time infrastructure.
 - Trained and deployed custom wake-word detection models, optimizing streaming connection latency to achieve near-instantaneous conversational responsiveness.

- Bridged continuous audio streaming with backend AI processing pipelines, enabling seamless and natural user interactions.

- **Project Syndra — Social Media Automation Engine**

Python, Node.js, Docker, Web Scraping APIs

- Developed an end-to-end automated news aggregation and publishing engine tailored for social media platforms using Python and Node.js.
- Built a robust data pipeline to autonomously scrape, dynamically rank, and filter visual news content based on custom relevance algorithms.
- Integrated external APIs to schedule and publish high-quality visual content, completely automating the content pipeline and saving 20 hours of manual curation weekly.

- **Cortex — LLM Reasoning Visualization Tool**

React, TypeScript, Python, FastAPI, WebSockets

- Designed and deployed a full-stack software system dedicated to tracking, logging, and visually mapping the internal reasoning chains of Large Language Models (LLMs).
- Built a highly responsive frontend dashboard and robust backend architecture to process and display complex AI inference steps in real-time.
- Enabled developers to effectively debug, audit, and optimize LLM prompts and agent behaviors through comprehensive visual analytics.

- **ScaleApp — Property Portfolio & Financial Dashboard**

PostgreSQL, Prisma, Python, LLM Prompting

- Spearheaded the development and refactoring of a comprehensive property portfolio management dashboard and data integration platform.
- Engineered complex data models and seamless pipelines to centralize real estate metrics using PostgreSQL.
- Designed and implemented structured system prompts for LLMs to generate automated, data-driven financial strategy interpretations for portfolio optimization.